
A Machine Learning System for EMG-Based Hand Gesture Recognition Using Random Forest Classification

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Abstract

Surface electromyography (sEMG) signals capture electrical muscle activity and can be used to infer human hand gestures for applications such as prosthetics, human–computer interaction, and rehabilitation interfaces. This project develops an end-to-end machine learning system for gesture recognition using the NinaPro DB1 dataset. The system includes data preprocessing, window segmentation, handcrafted feature extraction, and Random Forest classification. With a 50-dimensional feature vector extracted from each EMG window, the model achieves approximately 78% accuracy across 12 hand gesture classes. The pipeline is implemented with an emphasis on clarity, reproducibility, and systems-level reasoning rather than model complexity. This report describes the motivation, technical design, experiments, results, limitations, and potential extensions.

1. Introduction

Electromyography (EMG) signals measure muscle activation non-invasively and are widely used for robotic prosthetics, motor rehabilitation, and gesture-controlled systems. Despite their utility, EMG signals are noisy, highly variable across users, and sensitive to electrode placement. Machine learning provides a natural approach to mapping EMG segments to discrete hand gestures.

The goal of this project is to build a complete machine learning system—from raw physiological signals to final classification—focusing on (i) reproducible preprocessing, (ii) clear pipeline design, and (iii) classical ML techniques that perform well with limited engineering overhead. This work uses the NinaPro DB1 dataset and evaluates a Random For-

est classifier trained on well-established EMG time-domain features.

2. Related Work

Traditional EMG classification uses handcrafted features such as Mean Absolute Value (MAV), Root Mean Square (RMS), Zero Crossings (ZC), Slope Sign Changes (SSC), Waveform Length (WL), and autoregressive (AR) coefficients. These features capture both amplitude and frequency-related properties of EMG signals.

More recent methods utilize deep learning architectures (CNNs, LSTMs, Transformers) to automatically extract features from raw EMG. While these models can outperform classical approaches, they often require larger datasets, subject-specific calibration, and more computational resources.

Given the course emphasis on systems thinking, reproducibility, and practical pipeline design, this project intentionally uses classical ML instead of deep learning.

3. Data

3.1. Dataset

The NinaPro DB1 Exercise A1 dataset provides:

- 27 subjects,
- 10 surface EMG channels,
- 12 labeled gestures (static postures),
- Sampling rate: 100 Hz.

Each data file contains raw EMG signals and a corresponding *stimulus* signal encoding gesture labels.

3.2. Preprocessing

Raw EMG signals were segmented using sliding windows:

- window size: 400 samples,
- stride: 200 samples (50% overlap),

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- rest class (0) removed,
- per-subject normalization applied.

The final dataset contains 31,053 labeled windows.

4. Methods

4.1. Feature Extraction

For each of the 10 channels, the following features were computed:

- Mean Absolute Value (MAV),
- Root Mean Square (RMS),
- Waveform Length (WL),
- Zero Crossings (ZC),
- Slope Sign Changes (SSC),
- Autoregressive Coefficients (AR, order 4).

This produces a 50-dimensional feature vector per window.

4.2. Model: Random Forest

A Random Forest model was chosen for its robustness to noise and suitability for mixed-scale features. Hyperparameters:

```
n_estimators = 300
max_depth = None
min_samples_split = 2
n_jobs = -1
random_state = 42
```

The model was trained using an 80/20 stratified split.

5. Experiments and Results

5.1. Evaluation Metrics

Metrics used include:

- overall accuracy,
- precision, recall, and F1-score,
- confusion matrix (raw and normalized).

5.2. Results

The model achieves:

Test Accuracy: 78%

Common misclassifications occur between physiologically similar gestures.

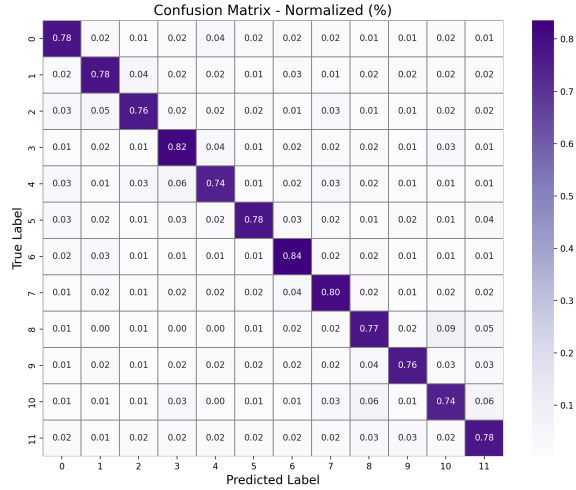


Figure 1. Normalized confusion matrix for the Random Forest classifier.

5.3. Interpretation

The confusion matrix reveals clusters of gestures that activate overlapping muscle groups. Random Forests handle EMG noise well, but subtle finger variations remain challenging.

6. Discussion

6.1. Strengths

- Pipeline is simple, interpretable, and efficient.
- Time-domain features offer strong signal representation.
- Random Forests provide robustness to noise.
- Implementation is reproducible and modular.

6.2. Limitations

- Cross-subject generalization is limited.
- Window size (400 samples) introduces latency for real-time use.
- Similar gestures remain hard to distinguish.
- No hyperparameter search was performed.

6.3. Future Work

Future improvements could include:

- subject adaptation,

- domain-adversarial training,
- smaller window sizes for real-time control,
- hybrid CNN+feature models.

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7. Conclusion

This project demonstrates an end-to-end EMG-based gesture recognition system using classical machine learning. With a well-designed feature extraction pipeline and Random Forest classifier, the system achieves strong accuracy and produces interpretable results. The project highlights the importance of careful preprocessing and systems organization in physiological ML applications.

Impact Statement

This work aims to advance machine learning for physiological signal processing. Potential applications include prosthetic limb control, rehabilitation tools, and assistive human–computer interfaces. While EMG-based systems raise no direct ethical risks beyond standard ML considerations, deployment must consider user variability, accessibility, and clinical reliability.

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